

A COUNTEREXAMPLE FOR [65] TO AN EXHAUSTIVENESS CONJECTURE OF ESPINOSA-GARCÍA AND PELLICER

ABSTRACT. We exhibit a balanced partition of [65] into seven Sidon sets that is not among the five partitions displayed by Espinosa-García and Pellicer or their reflections. This refutes their exhaustiveness conjecture in arXiv:2309.08553v1; reflection gives a second distinct omitted partition.

Write $[n] = \{1, \dots, n\}$. A finite set $A \subset \mathbb{Z}$ is *Sidon* if every integer has at most one representation $a + b$ with $a, b \in A$ and $a \leq b$. A partition of $[n]$ into Sidon sets is a *Sidon–Ramsey partition*; it is *balanced* if any two part sizes differ by at most one.

Espinosa-García and Pellicer display five balanced partitions of [65] into seven Sidon sets and then write, in unnumbered prose immediately after the displays [1, arXiv v1, p. 6]:

These partitions have two parts of 10 elements each, and five parts with 9 elements each. We conjecture that these and their reflected partitions (constructed by including the numbers $66 - x$ instead of x in each part) are all balanced Sidon-Ramsey partitions with those parameters.

We call this their *exhaustiveness conjecture*; it carries no number in [1]. Since $65 = 7 \cdot 9 + 2$, every such partition has two parts of size 10 and five of size 9, in agreement with the quoted sentence.

Theorem 1. *The exhaustiveness conjecture quoted above is false.*

Proof. Let $\mathcal{P} = \{A_1, \dots, A_7\}$, where

$$\begin{aligned} A_1 &= \{1, 5, 13, 23, 26, 28, 37, 56, 57, 63\}, & A_2 &= \{2, 20, 21, 24, 33, 44, 50, 58, 60, 65\}, \\ A_3 &= \{3, 4, 27, 30, 32, 39, 43, 49, 64\}, & A_4 &= \{6, 15, 16, 22, 36, 40, 48, 51, 53\}, \\ A_5 &= \{7, 14, 19, 25, 29, 38, 52, 54, 55\}, & A_6 &= \{8, 11, 17, 34, 41, 45, 46, 59, 61\}, \\ A_7 &= \{9, 10, 12, 18, 31, 35, 42, 47, 62\}. \end{aligned}$$

The listed integers are pairwise distinct and have union [65]; the part sizes are 10, 10, 9, 9, 9, 9, 9.

For finite $A \subset \mathbb{Z}$, put

$$\Sigma(A) = \{a + b : a, b \in A, a \leq b\}.$$

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Then A is Sidon exactly when $|\Sigma(A)| = \binom{|A|+1}{2}$. Exact addition gives

i	1	2	3	4	5	6	7
$ A_i $	10	10	9	9	9	9	9
$ \Sigma(A_i) $	55	55	45	45	45	45	45

so every A_i is Sidon and $\mathcal{P}$ is a balanced Sidon–Ramsey partition of [65]. Equivalently, A is Sidon precisely when the $\binom{|A|}{2}$ differences $b - a$ with $a < b$ in A are pairwise distinct. That is a check on 45 differences for each of A_1 and A_2 and on 36 for each of the remaining five parts, 270 in all, and in each part these differences are pairwise distinct.

We regard partitions as unordered families of parts. The ten 10-element parts occurring in the five partitions displayed in [1, arXiv v1, p. 6] are the eight distinct sets

$$\begin{aligned} &\{1, 3, 15, 22, 30, 33, 46, 50, 55, 56\}, & \{2, 4, 16, 23, 31, 34, 47, 51, 56, 57\}, \\ &\{2, 9, 17, 21, 26, 27, 47, 49, 60, 63\}, & \{2, 6, 14, 24, 27, 29, 38, 57, 58, 64\}, \\ &\{7, 10, 15, 19, 33, 43, 44, 50, 63, 65\}, & \{3, 9, 10, 29, 38, 40, 43, 53, 61, 65\}, \\ &\{3, 5, 17, 24, 32, 35, 48, 52, 57, 58\}, & \{2, 6, 14, 16, 19, 37, 38, 44, 53, 64\}, \end{aligned}$$

of which the first two occur twice: the first is a part of the first and of the second displayed partition, the second a part of the third and of the fourth. None of the eight contains $\{2, 65\}$ and none contains $\{1, 64\}$. Therefore no 10-element part in those partitions or their reflections contains $\{2, 65\}$. Since $\{2, 65\} \subset A_2$, the partition $\mathcal{P}$ is not in the conjectured family.

Let $\rho(x) = 66 - x$ and apply ρ elementwise to sets and partitions. Reflection preserves sizes and the Sidon property, and the conjectured family is ρ -invariant. Hence $\rho(\mathcal{P})$ is another omitted partition. It is distinct from $\mathcal{P}$: the 10-element part $\rho(A_2)$ contains $\{1, 64\}$, whereas neither 10-element part of $\mathcal{P}$ does. $\square$

The result concerns precisely the statement quoted above, as it appears on p. 6 of arXiv:2309.08553v1; no complete census of the balanced Sidon–Ramsey partitions of [65] with seven parts is asserted.

We have not been able to consult the published version listed in [1], and the e-print stands unrevised at v1, so whether the quoted sentence appears unchanged there is left open. The mathematical content does not depend on this: $\mathcal{P}$ and $\rho(\mathcal{P})$ are balanced Sidon–Ramsey partitions of [65] into seven Sidon sets, and by the argument above neither is one of the five partitions displayed in [1] nor a reflection of one.

REFERENCES

- [1] M. A. Espinosa-García and D. Pellicer, *Update on Sidon–Ramsey numbers*, arXiv:2309.08553v1 [math.CO] (2023); published in *Discrete Appl. Math.* **378** (2026), 120–124, doi:10.1016/j.dam.2025.07.002.